# Digital machining database for machine learning
# Dataset 1 description

The purpose of this open-source database is to support machine learning algorithm development for machining process modeling. The database is made available to the research community and provides organized data for algorithm training and benchmarking. The data are generated using analytical (frequency domain) and numerical (time domain) simulation[1] for various milling conditions.

[1]Schmitz, T. and Smith, K.S., Machining Dynamics: Frequency Response to Improved Productivity, Second Edition, Springer, New York, NY, 2019.

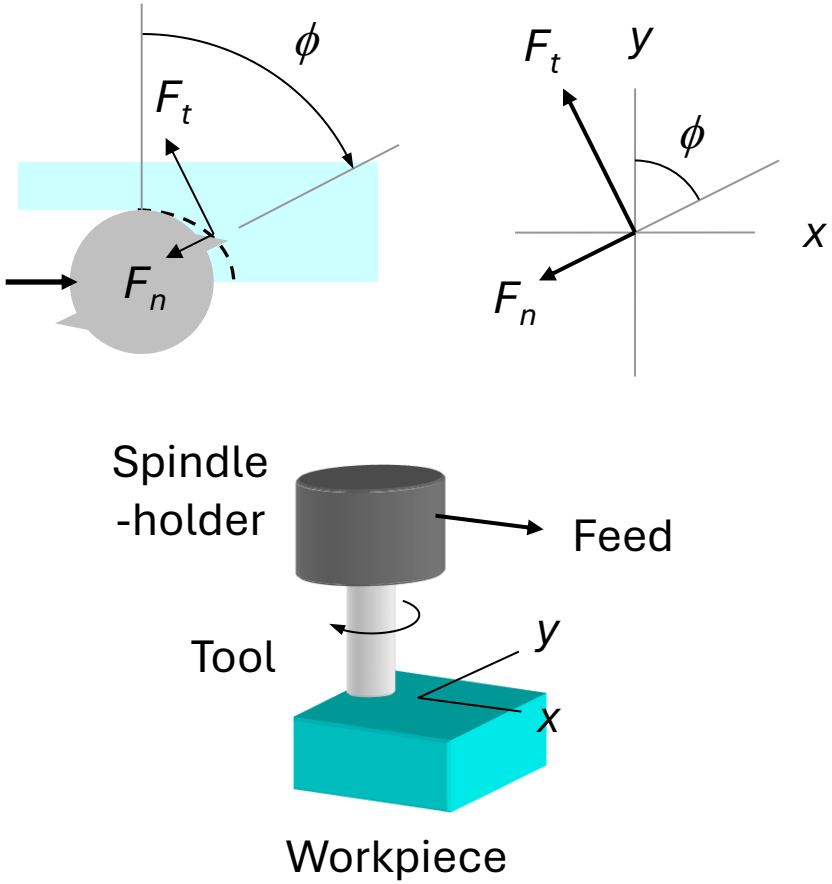
**Inputs**

Tool tip frequency response function modal parameters for the *x*, or feed, direction, where *z* is the endmill axis

- Each input is a vector to accommodate multiple vibration modes
  - Natural frequency, *fntx* = 500 Hz
  - Modal stiffness, *ktx* = 5e6 N/m
  - Modal viscous damping ratio, *zetatx* = 0.03

Tool tip frequency response function modal parameters for the *y* direction

- Each input is a vector to accommodate multiple vibration modes
  - Natural frequency, *fnty* = 500 Hz
  - Modal stiffness, *kty* = 5e6 N/m
  - Modal viscous damping ratio, *zetaty* = 0.03



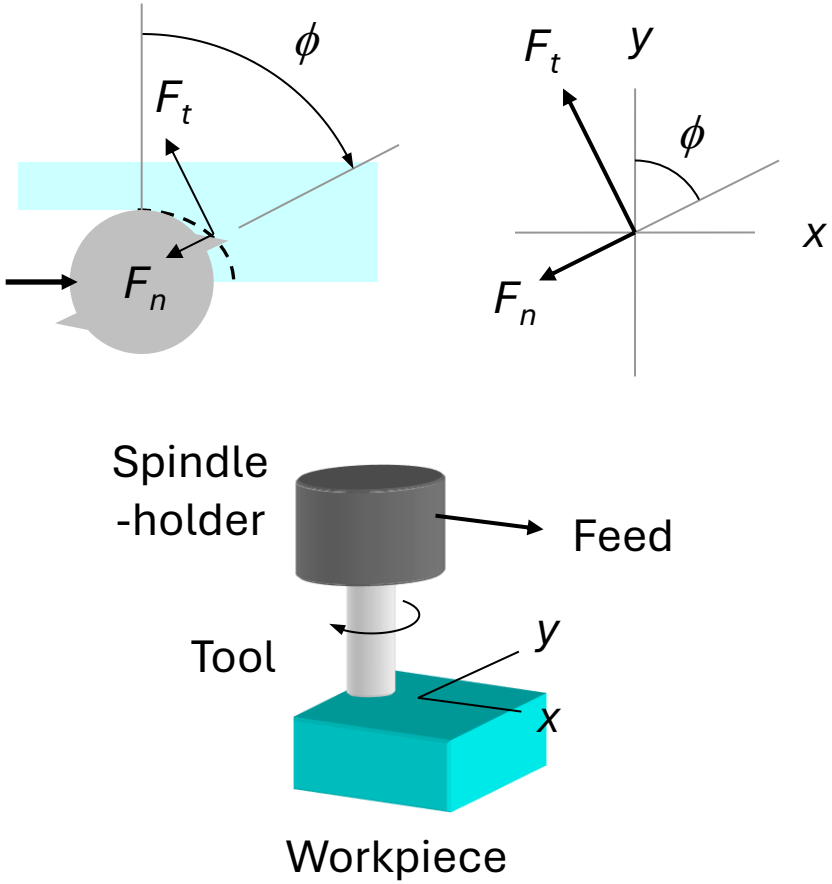
**Inputs**

Workpiece frequency response function modal parameters for the *x* direction

- Each input is a vector to accommodate multiple vibration modes
  - Natural frequency, *fnwx* = 500 Hz
  - Modal stiffness, *kwx* = 5e9 N/m
  - Modal viscous damping ratio, *zetawx* = 0.1

Workpiece frequency response function modal parameters for the *y* direction

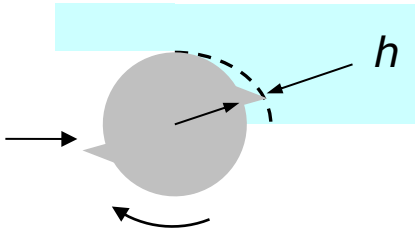
- Each input is a vector to accommodate multiple vibration modes
  - Natural frequency, *fnwy* = 500 Hz
  - Modal stiffness, *kwy* = 5e9 N/m
  - Modal viscous damping ratio, *zetawy* = 0.1

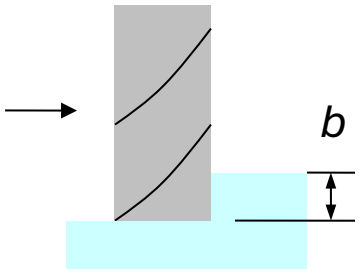


**Inputs**

Mechanistic cutting force model coefficients

- Tangential direction (*t*) cutting (*c*) coefficient, *ktc* = 700e6 N/m$^2$
- Normal direction (*n*) cutting coefficient, *knc* = 210e6 N/m$^2$
- Tangential direction edge (*e*) coefficient, *kte* = 0 N/m
- Normal direction edge coefficient, *kne* = 0 N/m
- Normal direction process damping coefficient, *Cn* = 0 N/m




Axial depth

Chip thickness

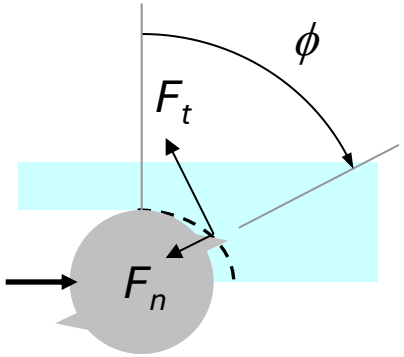
Tangential force $F_t = k_{tc} b h + k_{te} b$

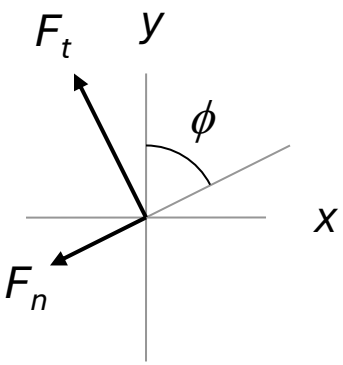
Normal force $F_n = k_{nc} b h + k_{ne} b$

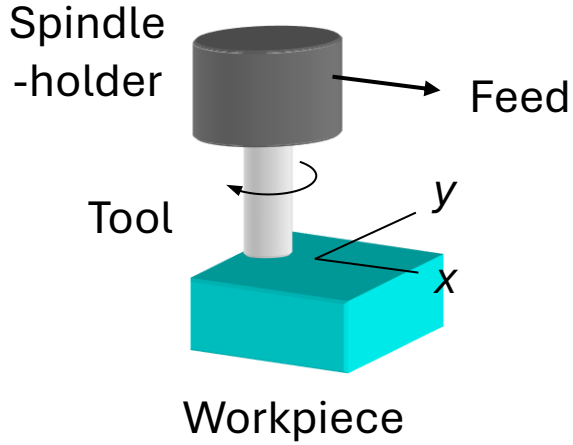
Process damping force $F_d = -C_n \frac{b}{v} \dot{n}$

Cutting speed

Normal direction velocity



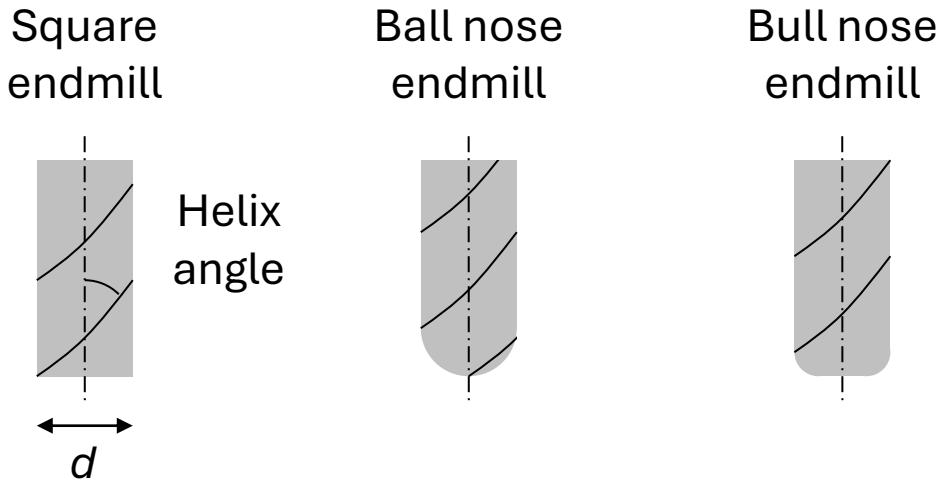



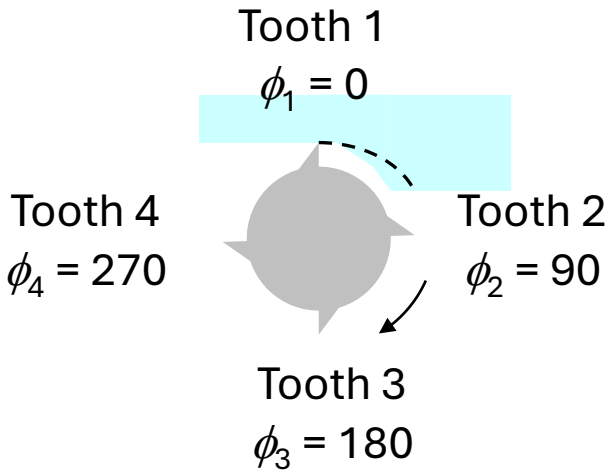



**Inputs**

Tool description

- Diameter, *d* = 12.7e-3 m
- Number of teeth, *m* = 4
- Helix angle vector with *m* entries, *beta* = [30 30 30 30] deg
- Tooth angle vector with *m* entries, *tooth_angle* = [0 90 180 270] deg
- Runout vector with *m* entries, *RO* = [0 0 0 0] m
- Endmill type: *et* = 1 for square



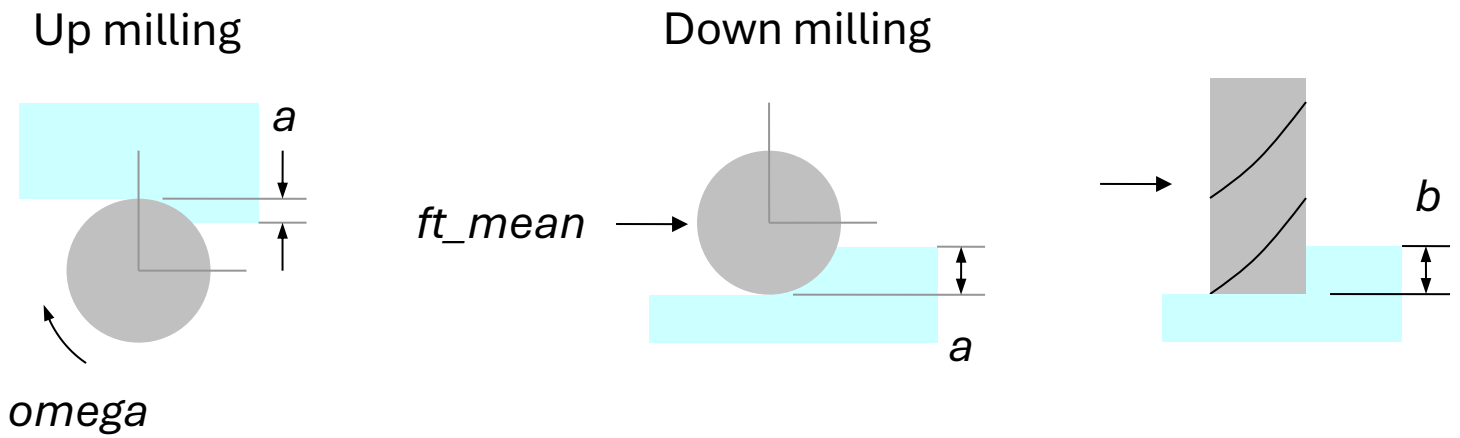




Example for uniform tooth spacing (not required) with four teeth

**Inputs**

Cut description

- Direction, *ud* = 2 for down (climb) milling
- Radial depth, *a* = 3.175e-3 m
- Axial depth, *b* = 1e-3 m to 15e-3 m in 1e-3 m increments (note: the grid is not fully populated; approximately two unstable results are included at each spindle speed only)
- Commanded feed per tooth, *ft_mean* = 0.1e-3 m
- Spindle speed, *omega* = 2000 rpm to 12000 rpm in 500 rpm increments

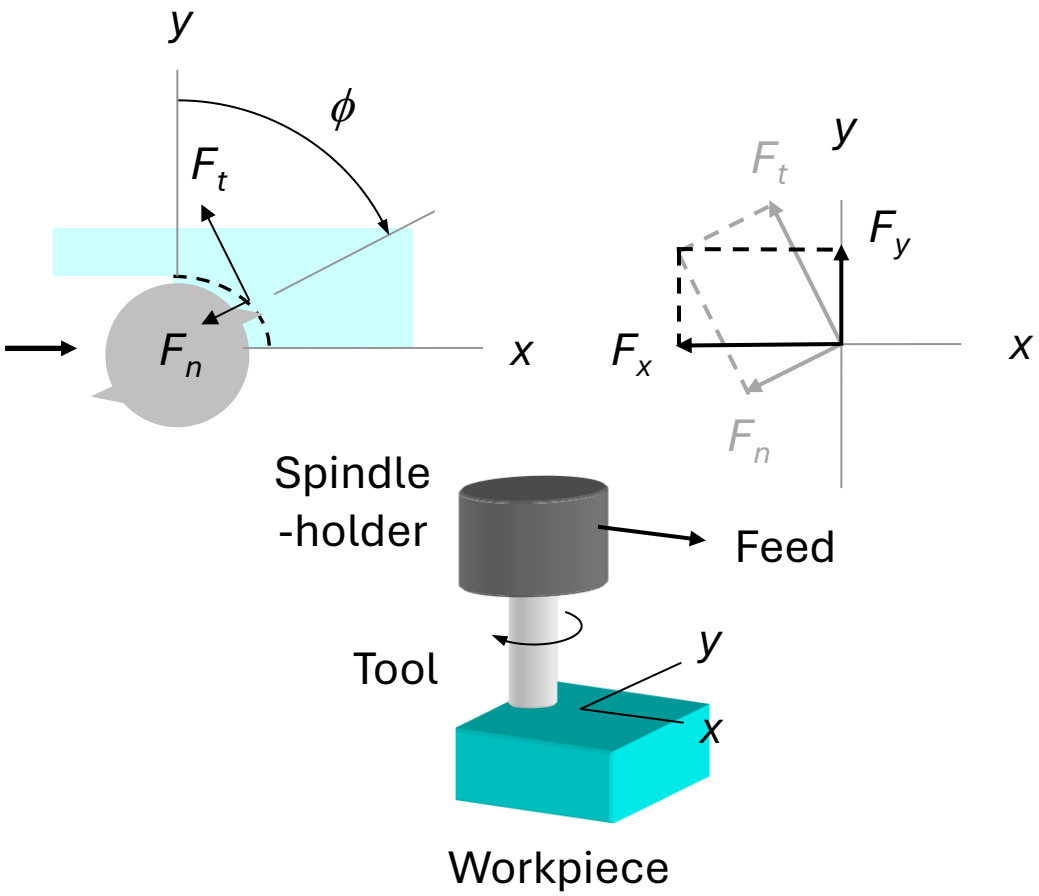


**Outputs**

Data files time1_y.mat and time1_y.h5 include

- Spindle speed, *omega* (rpm)
- Axial depth of cut, *b* (m)
- Time-based vectors
  - Discretized time, *time* (s)
  - Tool displacement in the *x* direction, *xtpos* (m)
  - Tool displacement in the *y* direction, *ytpos* (m)
  - Workpiece displacement in the *x* direction, *xwpos* (m)
  - Workpiece displacement in the *y* direction, *ywpos* (m)
  - Tool velocity in the *x* direction, *xtvel* (m/s)
  - Tool velocity in the *y* direction, *ytvel* (m/s)
  - Workpiece velocity in the *x* direction, *xwvel* (m/s)
  - Workpiece velocity in the *y* direction, *ywvel* (m/s)
  - Tool acceleration in the *x* direction, *xtacc* ($m/s^2$)
  - Tool acceleration in the *y* direction, *ytacc* ($m/s^2$)
  - Workpiece acceleration in the *x* direction, *xwacc* ($m/s^2$)
  - Workpiece acceleration in the *y* direction, *ywacc* ($m/s^2$)
  - Tool cutting force in the *x* direction, *Forcex* (N)
  - Tool cutting force in the *y* direction, *Forcey* (N)

In the data file names, 1 is the dataset number and y is the grid point number. The grid point numbers are ordered from the lowest spindle speed (lowest to highest axial depth at that speed) to the highest spindle speed (lowest to highest axial depth at that speed).

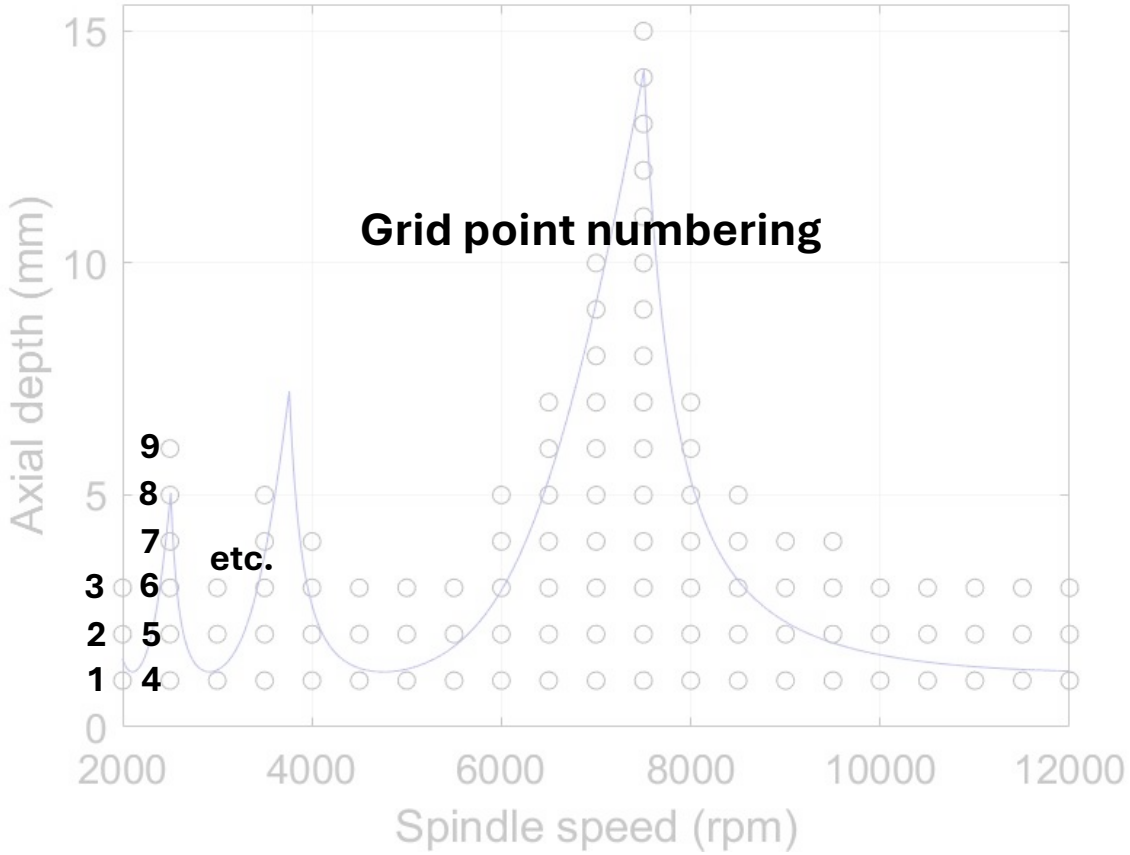


**Outputs**

Data files time1_y.mat and time1_y.h5 include

- Spindle speed, *omega* (rpm)
- Axial depth of cut, *b* (m)
- Time-based vectors
  - Discretized time, *time* (s)
  - Tool displacement in the *x* direction, *xtpos* (m)
  - Tool displacement in the *y* direction, *ytpos* (m)
  - Workpiece displacement in the *x* direction, *xwpos* (m
  - Workpiece displacement in the *y* direction, *ywpos* (m
  - Tool velocity in the *x* direction, *xtvel* (m/s)
  - Tool velocity in the *y* direction, *ytvel* (m/s)
  - Workpiece velocity in the *x* direction, *xwvel* (m/s)
  - Workpiece velocity in the *y* direction, *ywvel* (m/s)
  - Tool acceleration in the *x* direction, *xtacc* ($m/s^2$)
  - Tool acceleration in the *y* direction, *ytacc* ($m/s^2$)
  - Workpiece acceleration in the *x* direction, *xwacc* (m/s
  - Workpiece acceleration in the *y* direction, *ywacc* (m/s
  - Tool cutting force in the *x* direction, *Forcex* (N)
  - Tool cutting force in the *y* direction, *Forcey* (N)

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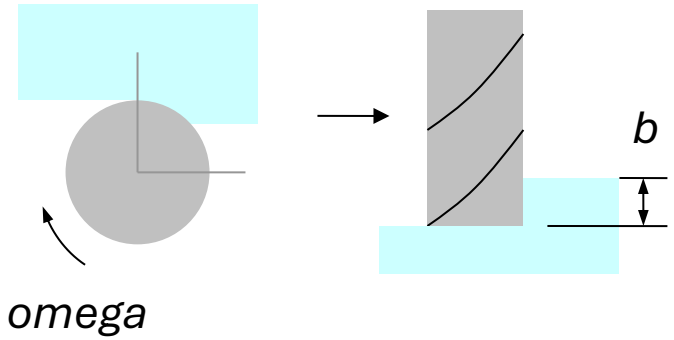


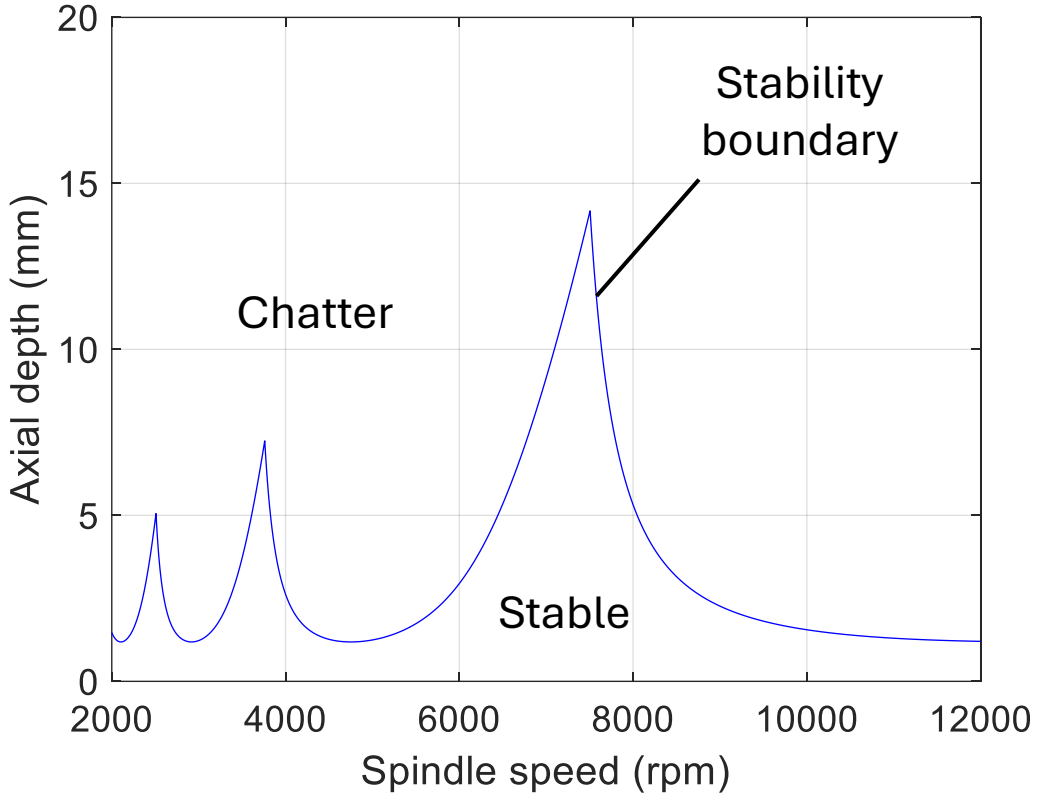
**Outputs**

Data files stability_boundary1.mat and stability_boundary1.h5 include

- Stability boundary vectors (Fourier series algorithm[1])
  - Spindle speed, *omega_vector* (rpm)
  - Limiting axial depth, *blim_vector* (m)

In the data file name, 1 is the dataset number.



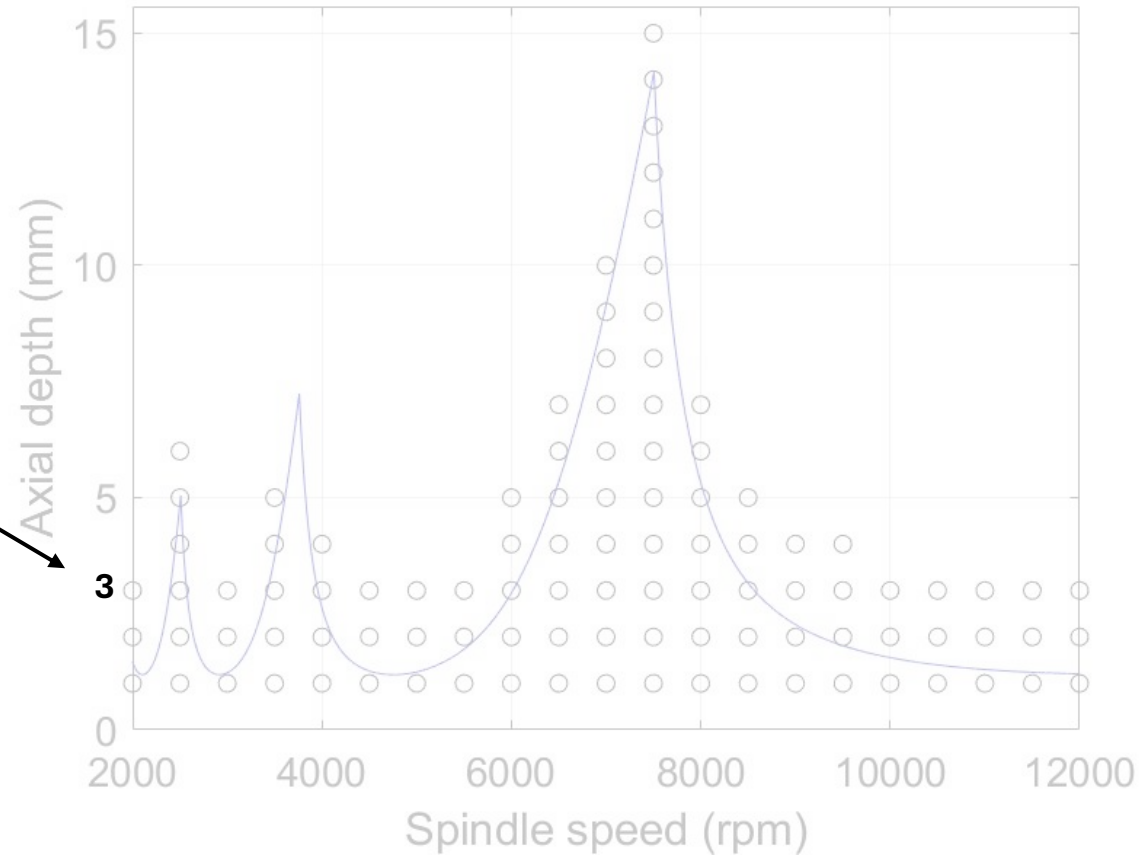

[1]Altintaş, Y. and Budak, E., 1995. Analytical prediction of stability lobes in milling. CIRP annals, 44(1), pp.357-362.

**Example Matlab code to open and plot data files**

**.mat time file**

```
load time1_3
figure(1)
subplot(211)
plot(time, xtpos*1e6, 'b')
set(gca,'FontSize', 14)
xlim([0 max(time)])
ylabel('x_t (\mum)')
grid on
zoom on
subplot(212)
plot(time, Forcex, 'b')
set(gca,'FontSize', 14)
xlim([0 max(time)])
xlabel('t (s)')
ylabel('F_x (N)')
grid on
zoom on
```

File 3 for
test point 3

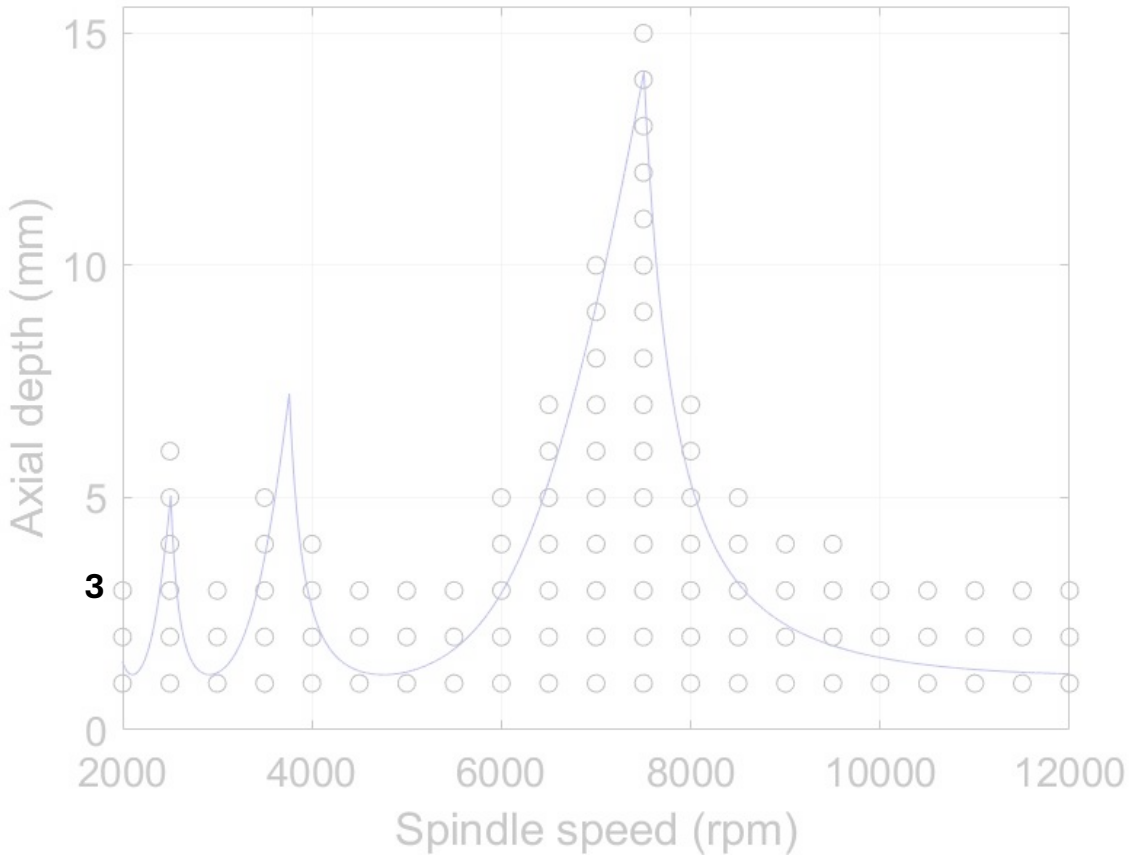


**Example Matlab code to open and plot data files**

**.h5 time file**

```
data = h5read("time1_3.h5", "/time1_3");
omega = data(1,1);
b = data(2,1);
time = data(3,:);
xtpos = data(4,:);
ytpos = data(5,:);
xwpos = data(6,:);
ywpos = data(7,:);
xtvel = data(8,:);
ytvel = data(9,:);
xwvel = data(10,:);
ywvel = data(11,:);
xtacc = data(12,:);
ytacc = data(13,:);
xwacc = data(14,:);
ywacc = data(15,:);
Forcex = data(16,:);
Forcey = data(17,:);
```

```
figure(1)
subplot(211)
plot(time, xtpos*1e6, 'b')
set(gca,'FontSize', 14)
xlim([0 max(time)])
ylabel('x_t (\mum)')
grid on
zoom on
subplot(212)
plot(time, Forcex, 'b')
set(gca,'FontSize', 14)
xlim([0 max(time)])
xlabel('t (s)')
ylabel('F_x (N)')
grid on
zoom on
```



**Example Matlab code to open and plot data files**

**.mat stability file**

```
load stability_boundary1
figure(1)
plot(omega_vector, blim_vector*1e3, 'b')
set(gca,'FontSize', 14)
axis([min(omega_vector) max(omega_vector) 0 max(blim_vector)*1.1e3])
xlabel('Spindle speed (rpm)')
ylabel('Axial depth (mm)')
grid on
zoom on
hold on
```

**Example Matlab code to open and plot data files**

**.h5 stability file**

```
data = h5read("stability_boundary1.h5", "/stability_boundary1");
omega_vector = data(1,:);
blim_vector = data(2,:);
figure(1)
plot(omega_vector, blim_vector*1e3, 'b')
set(gca,'FontSize', 14)
axis([min(omega_vector) max(omega_vector) 0 max(blim_vector)*1.1e3])
xlabel('Spindle speed (rpm)')
ylabel('Axial depth (mm)')
grid on
zoom on
```